

and do rule out order effects on accuracy based on the first two designs.

3.2.4 Participant Recruitment

Participants for the first and third designs, which constitute the majority, were recruited through teleradiology companies. The teleradiology companies allow us to recruit several experts in a relatively liquid spot market, a practice that is now common for decision experiments with non-expert subjects (Hunt and Scheetz, 2019). Most healthcare providers in the US rely on these companies’ services, although many large hospitals have on-call radiologists (Rosenkrantz et al., 2019). We work with teleradiology companies that serve US hospitals and offer the services of both US-based and non-US-based radiologists. Our contracts with teleradiology companies specify a piece-rate, and the companies, in turn, compensate the participants with a piece-rate.¹³ In addition, we provided monetary incentives for accuracy to a subset of radiologists, as described in the next section.

The second design required us to work with a partner who could guarantee subjects’ participation over several months. We collaborated with VinMac healthcare system in Vietnam to recruit their staff radiologists to ensure continued participation. VinMac is in the process of developing its own in-house AI capabilities and was willing to assist with our experiment in exchange for recognition in a publication of the resulting dataset. The VinMac radiologists did not receive any payments to participate in the experiment but we find that their performance is very close to the performance of the tele-radiologists.

In total, 180 radiologists participated in our experiment. Close to 17% of our participants are US-based, 38% have a degree from a US institution, 80% are affiliated with a large clinic, and 61% with an academic institution. As demonstrated in appendix C.3, the quality of the assessments made by the radiologists in our study is comparable to that of the staff radiologists from Stanford University Hospital, who originally diagnosed the patient case.

3.2.5 Incentives

We cross-randomize incentives for accuracy in the first and third designs but not the second because of the specific ways in which our partner’s radiologists are employed.¹⁴ Payments were determined following the binarized scoring rule in Hossain and Okui (2013), where truth

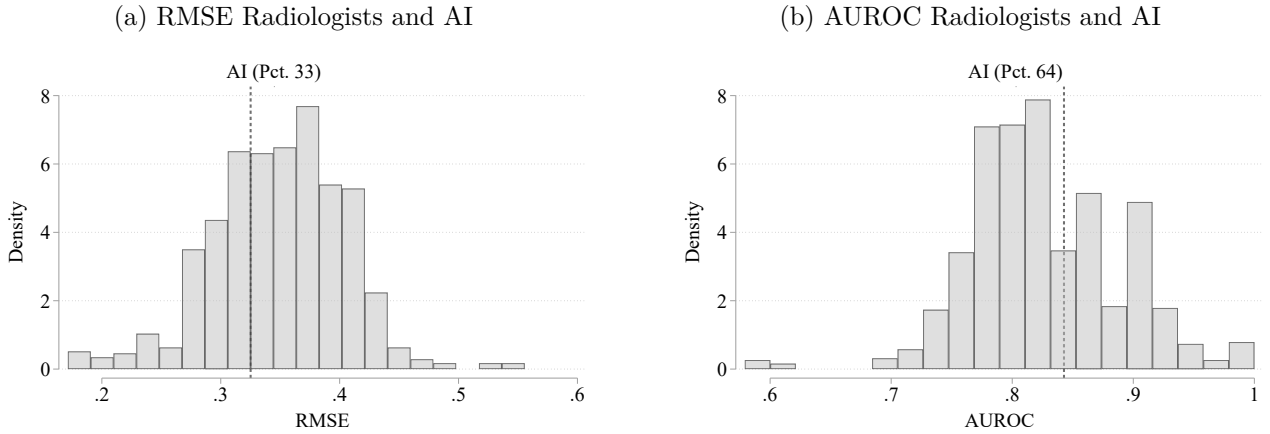
¹³We worked with three companies with piece-rates ranging from \$7.50 to \$13.00.

¹⁴There does not appear to be a consensus in the experimental literature on whether incentives are superior to non-incentivized responses when eliciting beliefs (see Danz, Vesterlund, and Wilson, 2020; Charness, Gneezy, and Rasocho, 2021). A comparison between the incentivized and non-incentivized participants allows us to answer this question in the context of this experiment.

lower RMSE indicates higher performance.²³ We pool the data for top-level pathologies with AI for each radiologist’s reports and for the AI’s prediction.

Figure 1 shows significant heterogeneity in performance across radiologists as well as the scope for AI assistance to improve radiologist performance. The AUROC shows that the bottom tail of the distribution of radiologists performs worse than the AI, whereas the upper tail predicts close to perfectly. This heterogeneity aligns with findings from observational data based on only treatment decisions (see Chan et al., 2022). The AI is more predictive than approximately two-thirds of radiologists, whether compared using AUROC or RMSE. Thus, there is ample room for AI assistance to improve the performance of radiologists. In fact, a majority of radiologists would do better on average by simply following the AI prediction. We present pathology-specific comparison results in appendix C.2.

Figure 1: Comparing AI performance to radiologists



Note: These histograms show distributions of two different accuracy measures of radiologist assessments alongside the AI’s accuracy. The left graph shows the distribution of the RMSE while the right shows the distribution of the average AUROC. Both distributions are shrunk to the grand mean using empirical Bayes. These measures are for each radiologist and include the top-level pathologies. The dotted line is the average measure of the AI algorithm for the corresponding distribution. Only the assessments where contextual history information is available for the radiologists but not the AI prediction are considered. Robustness by design and ground truth definition can be found in sections C.4.1 & C.4.2.

We also compare the performances of our participants and the radiologist who originally diagnosed each patient case in appendix C.3.²⁴ There is no discernible difference between

²³An important distinction between the two measures is that the AUROC is ordinal whereas RMSE is cardinal.

²⁴We classified the original free text radiology reports associated with each case as positive, negative, or uncertain for each pathology using the CheXbert algorithm described in Smit et al. (2020). To facilitate comparisons, we also discretized the probability assessments elicited during the experiment into these three categories. Then, we compared the accuracy of the original reads against the radiologists participating in the experiment.